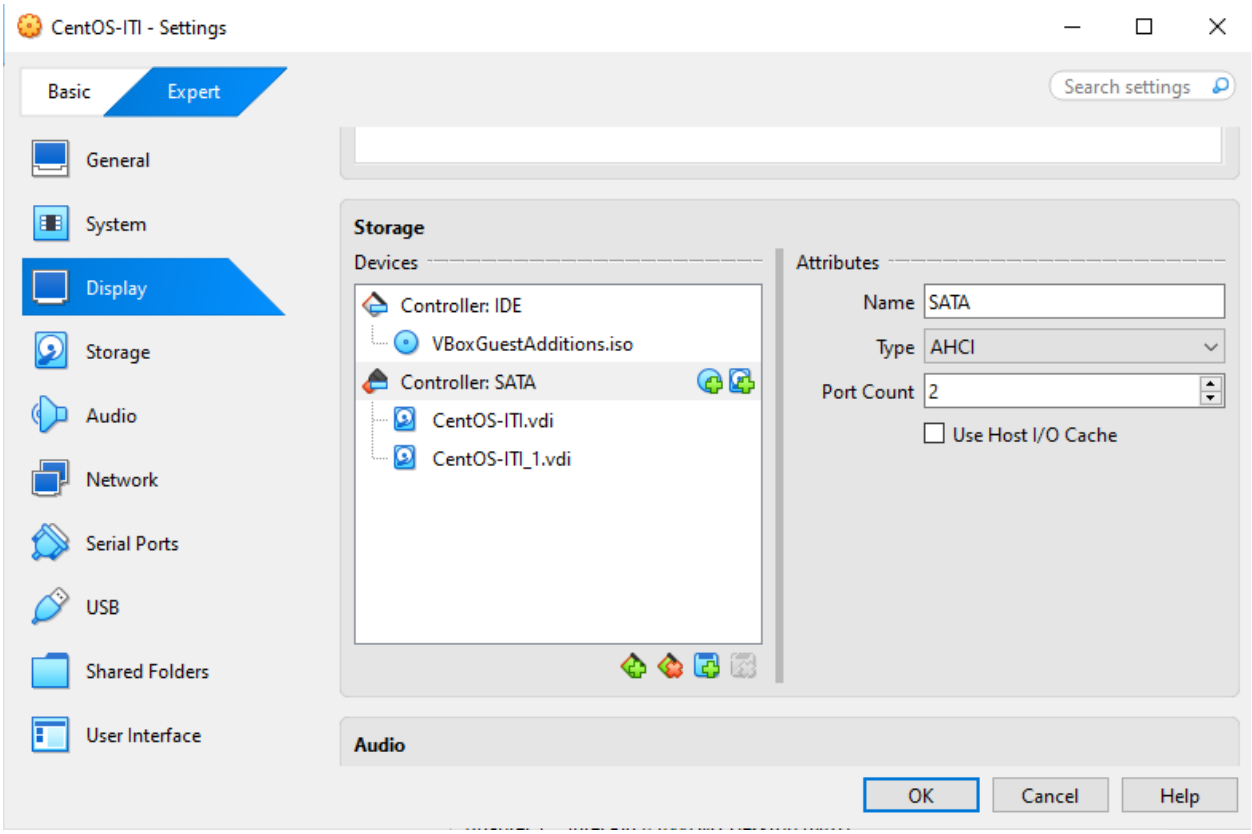
ITI – RedHat Admin2 lab5

Mohammad Tarek Omar – Data management

1. **Create and manage disk partitions using parted.**
2. **list all disks .**
3. **Start Partitioning.**
4. **Verify Partitions.**



```
(parted) mkpart primary ext4 0% 2GB
(parted) mkpart primary xfs 2GB 100%
(parted) print
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sdb: 5369MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name     Flags
 1      1049kB  2000MB  1999MB  ext4         primary
 2      2000MB  5368MB  3368MB  xfs          primary
```

```
[mohammadtarek@localhost ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   30G  0 disk
├sda1         8:1    0    1G  0 part /boot
└sda2         8:2    0   29G  0 part
  ├cs-root 253:0    0   27G  0 lvm  /
  └cs-swap 253:1    0    2G  0 lvm  [SWAP]
sdb           8:16   0    5G  0 disk
├sdb1         8:17   0  1.9G  0 part
└sdb2         8:18   0  3.1G  0 part
sr0          11:0    1 50.7M  0 rom  /run/media/mohammadtarek/VBox_GAs_7.2.4
[mohammadtarek@localhost ~]$
```

5. **Format the Partition(Add FileSystem).**

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo mkfs.ext4 /dev/sdb1
[sudo] password for mohammadtarek:
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 487936 4k blocks and 122160 inodes
Filesystem UUID: 4be182db-06d7-4c3c-84bb-60dcc729a77f
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[mohammadtarek@localhost ~]$ sudo mkfs.xfs /dev/sdb2
meta-data=/dev/sdb2              isize=512    agcount=4, agsize=205568 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data     =                       bsize=4096   blocks=822272, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% M
OUNTPOINTS
sda
├sda1
│    xfs                fa45b285-794c-458a-83c6-f1c4574b85d3    468.5M    51% /
boot
└sda2
     LVM2_m LVM2        2EliNz-qBWt-sXtb-0zdg-Gl62-n2oM-v1iB0N
  ├cs-root
  │  xfs                dba4bd05-0a12-4cdb-9292-593b48675a30     17.6G    34% /
  └cs-swap
     swap   1           7d171189-c933-4889-8cca-2329dd0c3df5                  [
SWAP]
sdb
├sdb1
│    ext4   1.0         4be182db-06d7-4c3c-84bb-60dcc729a77f
└sdb2
     xfs                286d3cbb-5bcf-4809-9f84-3d3f28caff4b
sr0  iso966 Jolie VBox_GAs_7.2.4
                        2025-10-17-10-20-48-34                       0   100% /
run/media/mohammadtarek/VBox_GAs_7.2.4
[mohammadtarek@localhost ~]$
```

6. **Mount the Partition.**

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo mkdir /mnt/data1
[mohammadtarek@localhost ~]$ sudo mkdir /mnt/data2
[mohammadtarek@localhost ~]$ sudo mount /dev/sdb1 /mnt/data1
[mohammadtarek@localhost ~]$ sudo mount /dev/sdb2 /mnt/data2
[mohammadtarek@localhost ~]$ df -h
Filesystem           Size  Used Avail Use% Mounted on
devtmpfs             4.0M     0  4.0M   0% /dev
tmpfs                854M     0  854M   0% /dev/shm
tmpfs                342M  5.5M  337M   2% /run
/dev/mapper/cs-root   27G  9.3G   18G  35% /
/dev/sda1            960M  492M  469M  52% /boot
tmpfs                171M  100K  171M   1% /run/user/1000
/dev/sr0              51M   51M     0 100% /run/media/mohammadtarek/VBox_GAs_7.
2.4
/dev/sdb1            1.8G   24K  1.7G   1% /mnt/data1
/dev/sdb2            3.1G   55M  3.1G   2% /mnt/data2
[mohammadtarek@localhost ~]$
```

for permenant mount even after restarting the device.

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="4be182db-06d7-4c3c-84bb-60dcc729a77f" TYPE="ext4" PARTLABEL="p
rimary" PARTUUID="4cdd54a2-74f1-4253-8b44-04c45598f1c4"
[mohammadtarek@localhost ~]$ sudo vi /etc/fstab
```

```
UUID=4be182db-06d7-4c3c-84bb-60dcc729a77f /mnt/data1  ext4  defaults  0 0
```

7. **Verify the Mount.**

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ df -h /mnt/data1
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       1.8G   24K  1.7G   1% /mnt/data1
[mohammadtarek@localhost ~]$
```

8. **Mounting File Systems Persistently(Configure file systems to mount automatically at boot) → Already made it on Q6 before looking on this question.**

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="4be182db-06d7-4c3c-84bb-60dcc729a77f" TYPE="ext4" PARTLABEL="p
rimary" PARTUUID="4cdd54a2-74f1-4253-8b44-04c45598f1c4"
[mohammadtarek@localhost ~]$ sudo vi /etc/fstab
```

```
UUID=4be182db-06d7-4c3c-84bb-60dcc729a77f /mnt/data1  ext4  defaults  0 0
```

9. **Create a SWAP Partition.**

```
[mohammadtarek@localhost ~]$ sudo umount /dev/sdb2
[mohammadtarek@localhost ~]$ sudo mkswap /dev/sdb2
mkswap: /dev/sdb2: warning: wiping old xfs signature.
Setting up swapspace version 1, size = 3.1 GiB (3368022016 bytes)
no label, UUID=ca2ba63c-6f74-450f-9030-b021cf22d661
```

10. **Enable the SWAP Partition.**

```
[mohammadtarek@localhost ~]$ sudo swapon /dev/sdb2
```

11. **Verify SWAP Space.**

```
[mohammadtarek@localhost ~]$ sudo swapon --show
NAME      TYPE      SIZE   USED PRIO
/dev/dm-1 partition   2G 378.3M   -2
/dev/sdb2 partition 3.1G     0B   -3
[mohammadtarek@localhost ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.7Gi       886Mi       656Mi       2.0Mi       331Mi       820Mi
Swap:         5.2Gi       378Mi       4.8Gi
[mohammadtarek@localhost ~]$
```

12. **Add SWAP to /etc/fstab.**

```
/dev/dbs2d      none    swap    defaults        0       0
~
```

13. **Create a SWAP File . → Secured it with chmod that no one can open it except root.**

```
[mohammadtarek@localhost ~]$ sudo dd if=/dev/zero of=/swapfile bs=1M count=1024
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 3.15059 s, 341 MB/s
[mohammadtarek@localhost ~]$ sudo chmod 600 /swapfile
```

14. **Enable the SWAP Partition.**

```
[mohammadtarek@localhost ~]$ sudo mkswap /swapfile
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=1f0ec554-9b8a-4ed2-ad49-090565aca8ff
[mohammadtarek@localhost ~]$ sudo vi /etc/fstab
[mohammadtarek@localhost ~]$
```

15. **Verify SWAP Space.**

```
[mohammadtarek@localhost ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.7Gi       788Mi        77Mi       2.0Mi       1.0Gi       919Mi
Swap:         5.2Gi       464Mi       4.7Gi
[mohammadtarek@localhost ~]$
```

16. **Add SWAP to /etc/fstab.**

```
/swapfile       none    swap    defaults        0       0
```

17. **Create Physical Volumes.**

```
[mohammadtarek@localhost ~]$ sudo pvcreate /dev/sdb2
  Physical volume "/dev/sdb2" successfully created.
```

18. **Verify the Physical Volume.**

```
[mohammadtarek@localhost ~]$ sudo pvs
  PV         VG Fmt  Attr PSize   PFree
  /dev/sda2  cs lvm2 a--  <29.00g     0
  /dev/sdb2     lvm2 ---   <3.14g <3.14g
```

19. **Create a Volume Group.**

```
[mohammadtarek@localhost ~]$ sudo vgcreate new_vg /dev/sdb2
  Volume group "new_vg" successfully created
```

20. **Verify the Volume Volume.**

```
[mohammadtarek@localhost ~]$ sudo vgs
  VG     #PV #LV #SN Attr   VSize   VFree
  cs       1   2   0 wz--n- <29.00g    0
  new_vg   1   0   0 wz--n-   3.13g 3.13g
```

21. **Create a Logical Volume.**

```
[mohammadtarek@localhost ~]$ sudo lvcreate -L 1G -n new_lv new_vg
  Logical volume "new_lv" created.
```

22. **Verify the Logical Volume.**

```
[mohammadtarek@localhost ~]$ sudo lvs
  LV     VG     Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Con
vert
  root   cs     -wi-ao---- 26.95g

  swap   cs     -wi-ao----  2.04g

  new_lv new_vg -wi-a-----  1.00g
```

23. **Format the Logical Volume.**

```
[mohammadtarek@localhost ~]$ sudo mkfs.ext4 /dev/new_vg/new_lv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: d23200bc-1903-4f28-9eb0-aeda7f6e2986
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

24. Mount the Logical Volume.

```
[mohammadtarek@localhost ~]$ sudo mkdir /mnt/storage_test
[mohammadtarek@localhost ~]$ sudo mount /dev/new_vg/new_lv /mnt/storage_test
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
```

25. Extend the Logical Volume.

```
[mohammadtarek@localhost ~]$ sudo lvextend -r -L +500M /dev/new_vg/new_lv
[sudo] password for mohammadtarek:
  File system ext4 found on new_vg/new_lv mounted at /mnt/storage_test.
  Size of logical volume new_vg/new_lv changed from 1.00 GiB (256 extents) to <1.
49 GiB (381 extents).
  Extending file system ext4 to <1.49 GiB (1598029824 bytes) on new_vg/new_lv...
resize2fs /dev/new_vg/new_lv
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/new_vg/new_lv is mounted on /mnt/storage_test; on-line resizin
g required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/new_vg/new_lv is now 390144 (4k) blocks long.

resize2fs done
  Extended file system ext4 on new_vg/new_lv.
  Logical volume new_vg/new_lv successfully resized.
[mohammadtarek@localhost ~]$
```

26. Verify the Logical Volume.

```
[mohammadtarek@localhost ~]$ df -h /mnt/storage_test
Filesystem                 Size  Used Avail Use% Mounted on
/dev/mapper/new_vg-new_lv  1.5G   24K  1.4G   1% /mnt/storage_test
[mohammadtarek@localhost ~]$
```

27. Reduce the Logical Volume.

```
[mohammadtarek@localhost ~]$ sudo lvreduce -r -L -200M /dev/new_vg/new_lv
  File system ext4 found on new_vg/new_lv mounted at /mnt/storage_test.
  File system size (<1.49 GiB) is larger than the requested size (1.29 GiB).
  File system reduce is required using resize2fs.
  File system unmount is needed for reduce.
  File system fsck will be run before reduce.
Continue with ext4 file system reduce steps: unmount, fsck, resize2fs? [y/n]:y
  Reducing file system ext4 to 1.29 GiB (1388314624 bytes) on new_vg/new_lv...
unmount /mnt/storage_test
unmount done
e2fsck /dev/new_vg/new_lv
/dev/new_vg/new_lv: 11/98304 files (0.0% non-contiguous), 15140/390144 blocks
e2fsck done
resize2fs /dev/new_vg/new_lv 1355776k
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/new_vg/new_lv to 338944 (4k) blocks.
The filesystem on /dev/new_vg/new_lv is now 338944 (4k) blocks long.

resize2fs done
remount /dev/new_vg/new_lv /mnt/storage_test
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
remount done
  Reduced file system ext4 on new_vg/new_lv.
```

28. Verify the Logical Volume.

```
[mohammadtarek@localhost ~]$ sudo lvs
  LV     VG     Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Con
vert
  root   cs     -wi-ao---- 26.95g

  swap   cs     -wi-ao----  2.04g

  new_lv new_vg -wi-ao----  1.29g

[mohammadtarek@localhost ~]$ df -h /mnt/storage_test
Filesystem                 Size  Used Avail Use% Mounted on
/dev/mapper/new_vg-new_lv  1.3G   24K  1.2G   1% /mnt/storage_test
[mohammadtarek@localhost ~]$
```